

Review

A Review of Algae-Based Carbon Capture, Utilization, and Storage (Algae-Based CCUS)

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Abstract: Excessive emissions of greenhouse gases, primarily carbon dioxide (CO₂), have garnered worldwide attention due to their significant environmental impacts. Carbon capture, utilization, and storage (CCUS) techniques have emerged as effective solutions to address CO₂ emissions. Recently, direct air capture (DAC) and bioenergy with carbon capture and storage (BECCS) have been advanced within the CCUS framework as negative emission technologies. BECCS, which involves cultivating biomass for energy production, then capturing and storing the resultant CO₂ emissions, offers cost advantages over DAC. Algae-based CCUS is integral to the BECCS framework, leveraging algae's biological processes to capture and sequester CO₂ while simultaneously contributing to energy production and potentially achieving net negative carbon emissions. Algae's high photosynthetic efficiency, rapid growth rates, and ability to grow in non-arable environments provide significant advantages over other BECCS methods. This comprehensive review explores recent innovations in algae-based CCUS technologies, focusing on the mechanisms of carbon capture, utilization, and storage through algae. It highlights advancements in algae cultivation for efficient carbon capture, algae-based biofuel production, and algae-based dual carbon storage materials, as well as key challenges that need to be addressed for further optimization. This review provides valuable insights into the potential of algae-based CCUS as a key component of global carbon reduction strategies.



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1. Introduction

Carbon dioxide (CO₂), one of the primary greenhouse gases, has been steadily increasing in the atmosphere, leading to significant adverse effects globally [1,2]. These effects include intensified global warming and frequent extreme weather events, which severely threaten human livelihoods, crop production, wildlife habitats, and the overall stability and functionality of ecosystems. Reducing and controlling CO₂ emissions to address the challenges posed by climate change has become a critical issue of international concern in recent years. In response, international climate policies such as the United Nations Framework Convention on Climate Change (UNFCCC) [3] and the Paris Agreement [4,5] have driven global efforts to mitigate excessive CO₂ emissions and tackle climate change.

To meet CO₂ reduction targets, replacing traditional fossil fuels with clean energy sources, such as hydrogen, nuclear power, wind energy, and solar power, is considered an effective strategy [6–11]. Fossil fuels produce CO₂ during extraction, refining, and transportation, and emit large amounts of CO₂ during combustion [12–15]. Given the extensive reliance on fossil fuels for electricity generation, transportation, and industrial processes, this results in substantial CO₂ emissions. In contrast, clean energy sources can eliminate these emissions [16]. However, the high costs and limited technological

infrastructure of clean energy technologies currently restrict their widespread adoption. Thus, fossil fuels remain the dominant global energy source.

In response to this challenge, carbon capture, utilization, and storage (CCUS) technologies have emerged as a key solution to address excessive CO₂ emissions from fossil fuel use [17,18]. CCUS involves capturing CO₂ from industrial processes or energy production, utilizing it to create valuable products or chemicals, or securely storing it in underground reservoirs or stable geological formations to prevent its release into the atmosphere. Globally, countries are actively advancing CCUS technologies with significant progress. China, the largest emitter of CO₂, has implemented supportive policies for CCUS technologies and projects since the start of its 13th Five-Year Plan in 2016, which has substantially accelerated their development [17]. Notable advancements in China include CO₂ capture, CO₂-enhanced oil recovery (EOR), CO₂-enhanced gas recovery (EGR), CO₂-enhanced coal bed methane (ECBM), and CO₂ saline aquifer storage [19–22]. The United States is also at the forefront of CCUS research and application, bolstered by strong governmental support. Policies such as the 45Q Tax Credit [23,24], the United States Energy Act of 2020 [25], the Infrastructure Investment and Jobs Act [26], and the Utilizing Significant Emissions with Innovative Technologies (USE IT) Act [27] reflect the U.S. government's commitment to addressing climate change and achieving emission reduction goals. These policies and economic incentives underscore the vital role of the U.S. in mitigating climate change and advancing global emission reduction efforts. India, following China and the United States, is the third-largest emitter of CO₂ globally, due to its rapid industrialization and growing energy demands [28]. To address these emissions, India has set ambitious goals in its Intended Nationally Determined Contribution (INDC), aiming to achieve 175,000 MW of renewable energy capacity by 2022 and to ensure that around 40% of its installed capacity comes from non-fossil fuel sources by 2030 [29]. While India's primary focus has been on transitioning to renewable energy and enhancing energy efficiency, there is growing interest in CCUS technologies. Recently, the government launched the National Mission on Carbon Sequestration and Enhanced Oil Recovery (EOR), which targets capturing CO₂ from industrial sources for enhanced oil recovery in mature oil fields. This initiative highlights India's strategy to incorporate CCUS technologies into its climate change mitigation efforts [30]. Moreover, India has partnered with several international organizations and is collaborating with other countries to develop and demonstrate CCUS technologies in the Indian context [31–33].

Although CCUS technologies have made significant progress in reducing CO₂ emissions, achieving net-zero emissions remains a challenge that requires continued technological advancements. Direct air carbon capture and storage (DACCS) and bioenergy with carbon capture and storage (BECCS) are both considered essential for achieving carbon neutrality due to their potential to generate negative emissions [34]. DACCS captures and stores low-concentration CO₂ directly from the atmosphere, a process that requires highly efficient adsorbents and large-scale infrastructure, resulting in relatively high costs [35,36]. In contrast, BECCS is more cost-effective, as it produces bioenergy from biomass, and the CO₂ released during bioenergy production and utilization is reabsorbed by growing biomass, forming a closed-loop system. Moreover, BECCS generates bioenergy, adding economic value while reducing carbon emissions. Due to its economic and sustainability advantages, BECCS is more favorable for widespread global adoption compared to DACCS.

Algae capture CO₂ through photosynthesis, which can be utilized in the commercial production of high-value products such as food, nutraceuticals, pharmaceuticals, and cosmetics. This type of algae-based carbon utilization is well-established globally, with notable implementation in regions such as the southwestern United States, Asia, and Australia [37]. Although these applications do not involve bioenergy production, they remain a key part of algae-based carbon utilization. Within the BECCS framework, algae offer distinct advantages over conventional feedstocks like crops, wood, and agricultural residues [38–40] for bioenergy production. Algae possess high photosynthetic efficiency and rapid growth rates, allowing for more effective carbon capture [41]. In addition

to CO₂, algae can absorb sulfur dioxide and nitrogen oxides, further contributing to emission reductions [42,43]. Algae can also thrive in environments that are unsuitable for traditional agriculture, such as polluted water, minimizing competition for arable land and freshwater [44–46]. Moreover, algae-based technologies feature lower CO₂ storage and operational costs, enhancing their feasibility for large-scale deployment. Consequently, algae-based CCUS research has garnered growing attention in recent years, and companies such as Accelergy Corporation, Global Algae Innovations, Neste, and MicroBio Engineering are at the forefront of advancing these technologies toward commercialization.

Several studies have examined the potential of algae in CCUS, addressing various processes and associated challenges. Paul et al. highlighted the role of macroalgae and microalgae in carbon capture, concluding that optimizing open and closed cultivation systems enhances CO₂ fixation efficiency [47]. Yang et al. demonstrated that microalgae could serve as a cost-effective and efficient solid fuel with improved combustion performance, while also discussing the potential for industrial applications and the challenges [48]. Singh et al. underscored the promise of hydrothermal carbonization as a method for converting wet algal biomass into high-energy biocoal, offering a cleaner alternative to fossil fuels that supports both carbon sequestration and waste management [49]. Bibi et al. focused on bioethanol production from algae, emphasizing its sustainability and environmental benefits, while noting advancements in cultivation, harvesting, and commercialization techniques [50]. Pathy et al. explored the mechanisms driving large-scale biohydrogen production from algae, summarizing strategies to enhance hydrogen yield and ensure economic viability [51]. Lee et al. introduced advancements in pyrolysis technologies, highlighting the potential of algae-based syngas as a competitive and viable energy source [52].

These studies typically focus on specific aspects of algae applications in CCUS. However, there are still a lack of comprehensive reviews on algae-based CCUS through both organic and inorganic pathways. This research aims to address this gap by providing a thorough review of algae-based CCUS in this context. It begins by outlining the fundamental characteristics of algae and highlighting their unique advantages in achieving CO₂ emission reduction targets. Given that algae's ability to capture CO₂ is critical for subsequent carbon utilization and storage, this research emphasizes the mechanisms, influencing factors, and recent developments in algae-based CO₂ capture. Subsequently, this study details the technologies for applying captured CO₂ in a circular bioeconomy, focusing on efficiently converting carbon into high-value bioenergy products, including solid biomass pellets, biocoal, biodiesel, bioethanol, biohydrogen, biogas, and syngas. Furthermore, this research introduces the dual carbon storage theory, based on the mechanisms of algae's photosynthesis and microbially induced calcium carbonate precipitation, along with innovative algae-based dual carbon storage materials representing significant advancements in long-term carbon storage. Through this exploration of technological progressions in algae-based CCUS, the study identifies the current challenges faced by these technologies and proposes directions for future improvements. By offering insights into the latest advancements in algae-based CCUS, this review aims to foster technological innovation that ensures the long-term sustainability of negative emission technologies, thereby laying a robust foundation for achieving net zero carbon emission targets.

2. Data Collection and Methodology

This review study was conducted through a systematic process of information collection, classification, analysis, and summarization. The process began with extensive literature searches in major academic databases, including Web of Science, Scopus and CNKI, focusing on peer-reviewed journal articles and conference proceedings to gather the latest technological advancements in the field of algae-based CCUS. Initially, a total of 6195 publications were retrieved from the database search. After screening abstracts according to inclusion criteria, 1600 publications were selected for full-text review. Finally, approximately 280 studies were included in this review. Additionally, supplementary information was obtained from publicly available resources, including reports from inter-

national organizations, government publications, and relevant company websites. The covered publications span several decades, from the 1960s to 2024. The collected information was then meticulously categorized into sections: an overview of algae, algae-based carbon capture, algae-based carbon utilization, and algae-based carbon storage. Each section provides an in-depth exploration of relevant theories and the latest technological developments. Notably, while algae are also widely applied in various sectors such as food, feed, chemicals, and cosmetics, this review specifically focuses on how CO₂ captured by algae can be converted into high-value bioenergy products within the BECCS framework. Finally, by systematically analyzing existing algae-based CCUS technologies, this study identifies the constraints and challenges these technologies face and proposes potential solutions and future directions for improvement.

3. Overview of Algae

Algae are diverse, photosynthetic organisms found in aquatic environments, with relatively simple structures, either multicellular or unicellular [47]. They are polyphyletic, encompassing both prokaryotic organisms, such as Cyanophyceae (blue-green algae), and eukaryotic organisms, including Chlorophyceae (green algae), Bacillariophyceae (diatoms), and Chrysophyceae [53]. Algae are considered foundational to Earth's ecosystems, providing energy and organic matter to maintain ecological balance and biodiversity.

The primary constituents of algae include proteins, lipids, carbohydrates, cellulose, vitamins, minerals, as well as specific compounds like algal polysaccharides and pigments [54–56]. The composition may vary among different species of algae. Lipids, carbohydrates, and proteins serve as the main building blocks of algal biomass and form the basis for its conversion into food, cosmetics, pharmaceuticals, and biofuels [57].

3.1. Classification of Algae

Algae are commonly classified into two basic types based on size and structure: macroalgae and microalgae [47,53,58]. Figure 1 presents their structures, while Table 1 provides a detailed comparison of their characteristics.

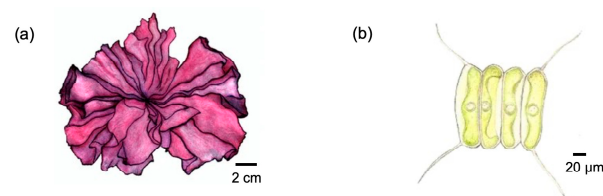


Figure 1. Schematic of algae structures: (a) structure of *Porphyra umbilicalis*, a representative of Rhodophyta in macroalgae; (b) structure of *Scenedesmus*, a representative of Chlorophyta in microalgae. Reproduced with permission from [58] under the CC BY license.

Table 1. Comparison of macroalgae and microalgae.

Algae Type	Characteristics	Growth Environment	Subcategories	Suitability for CCUS	References
Macroalgae	Complex structures composed of multicellular organisms; Low protein and lipid content but high carbohydrates and moisture	Coastal or shallow water areas, usually requiring substrate attachment	Classified into four groups based on pigmentation: Cyanophyta, Chlorophyta, Heterokontophyta, and Rhodophyta	High productivity; Bioremediation of contaminants; High conversion capability from inorganic carbon to biomass	[53,59,60]
Microalgae	Small-sized structures with single-celled or few-celled composition; Rich in proteins and lipids	Widely distributed in various aquatic environments	Classified into three groups based on pigmentation: Bacillariophyceae, Chlorophyceae, and Chrysophyceae	Rapid growth rate; High photosynthetic rate; High conversion efficiency	[53,60]

Macroalgae are large, multicellular organisms [58] that grow in coastal waters and require attachment to substrates [61,62]. Despite slower growth, they contribute significantly to carbon capture due to their large biomass. Microalgae, in contrast, are small,

single-celled organisms found in diverse aquatic environments [63,64], with rapid growth rates and high photosynthetic efficiency [65,66]. Despite their differences, both macroalgae and microalgae play significant roles in carbon capture, utilization, and storage.

3.2. Advantages of Algae in CCUS

The characteristics of algae, including their low environmental requirements for growth [67], high photosynthetic rates and short growth cycles [47], and their capability to produce high-value products [68,69], demonstrate their significant value in CCUS.

Algae cultivation offers significant advantages for CCUS applications due to its adaptability to various environmental conditions. Algae can grow in diverse water types, including freshwater [70,71], saltwater [72–74], and even wastewater [72]. In addition to macroalgae, which are commonly well-established in saltwater environments, certain microalgae, like *Dunaliella salina*, have been effectively cultivated in saltwater, reducing the need for freshwater [72]. This adaptability allows some microalgae to be cultivated on land unsuitable for traditional agriculture, such as non-arable or saline soils, thus avoiding competition with conventional crops for valuable arable land and freshwater resources. Additionally, algae can absorb and remove harmful substances from polluted water, contributing to water quality improvement during growing [44–46,75,76]. For instance, *Chlorella vulgaris* and *Scenedesmus obliquus* have been successfully cultivated in wastewater and have aided in the removal of pollutants like nitrogen and phosphorus [77,78]. *Spirulina platensis* has been shown to absorb and detoxify heavy metals in contaminated water while contributing to biomass production by capturing CO₂ [79].

Algae exhibit higher photosynthetic rates and shorter growth cycles compared to traditional terrestrial plants [47]. For example, microalgae such as *Chlorella vulgaris* can double their biomass within 3 days under optimal conditions [80], while macroalgae like *Ulva tephida* typically takes 2 to 3 weeks to reach a harvestable size [81]. In contrast, many terrestrial plants, such as wheat and corn, take months to reach maturity. The rapid growth of algae allows them to absorb CO₂ more efficiently, resulting in higher CO₂ capture rates. Currently, commercial algae cultivation can capture and sequester nearly two tons of CO₂ for every ton of algae produced [82], making it a highly effective method for carbon storage compared to terrestrial plants.

Algal biomass has diverse applications and can be effectively utilized to produce high-value products such as food [83], feed [84], chemicals and biochemicals [68], biobased materials [85–87], and biofuels [88]. In particular, macroalgae, such as kelp, Saccharina, and Ulva, are primarily used in the production of food, feed, and chemicals and biochemicals due to their rich nutritional qualities and bioactive compounds. In contrast, microalgae are commonly utilized for biofuel production due to their high biomass content and rapid growth rates. Among the biofuels derived from microalgae, biodiesel and bioethanol generally hold the highest potential due to well-established thermochemical processing methods and the energy-dense nature of their lipid and carbohydrate contents. Notably, the production of biofuels from microalgae exemplifies a sustainable closed-loop process. During biofuel combustion, CO₂ is emitted and can then be absorbed by algae for photosynthesis as part of a natural cycle, thus creating a continuous and self-sustaining carbon process.

Overall, algal systems, or more specifically algae biomass-derived circular systems, are particularly well-suited for CCUS applications. Algae effectively capture CO₂ through photosynthesis and convert it into biomass, which can then be used to produce biofuels. This process facilitates both the utilization and storage of CO₂, contributing to the achievement of emission reduction goals.

4. Algae-Based Carbon Capture

The ability of algae to capture and fix CO₂ is fundamental to its subsequent utilization and storage of carbon, primarily through the process of photosynthesis [89]. Under visible light, algae convert CO₂ and water into organic compounds while releasing oxygen [90],

as overall Reaction (1). This process involves two stages: light-dependent and light-independent reactions, with the associated reactions illustrated in Figure 2.

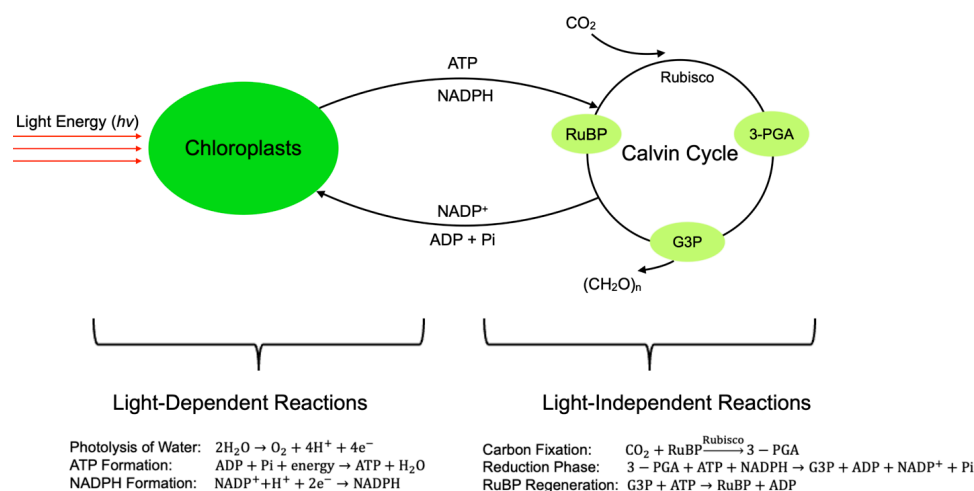
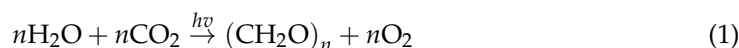


Figure 2. Schematic of the photosynthetic carbon fixation process in algae.

A specialized adaptation in algae for enhancing CO_2 fixation is the carbon-concentrating mechanism (CCM), which helps reduce photorespiration—a process that decreases the efficiency of CO_2 fixation when oxygen competes with CO_2 for Rubisco binding sites [47]. The CCM uses specific transport proteins to import bicarbonate ions (HCO_3^-) into algal cells, where carbonic anhydrase converts them into CO_2 , enhancing the intracellular CO_2 concentration [91]. This elevated CO_2 concentration inside the cell gives CO_2 a competitive advantage over O_2 for binding to Rubisco, improving the efficiency of CO_2 fixation through photosynthesis in algae [47].

4.1. Factors Influencing Algae-Based Carbon Capture

The carbon capture capacity of algae is primarily determined by factors including algal species, light conditions, temperature and pH, and the cultivation medium composition.

Algal species with high photosynthetic efficiency, rapid growth rates, high storage compound capacity, and strong environmental adaptability are key to effective carbon capture and biomass conversion in CCUS processes, as they enhance CO_2 fixation, biomass yield, and resilience under various conditions [92–94]. Light conditions, including intensity [95], distribution [96], penetration [97], and duration [95,98], must be optimized to prevent photoinhibition [99], which produces reactive oxygen species (ROS) [100] that can damage cellular structures and functions [101,102]. Additionally, sufficient light must reach deeper algal layers to maximize photosynthetic activity and carbon fixation throughout the entire cultivation system.

Most algae exhibit a maximum carbon capture and biomass production within a temperature range of $15\text{ }^\circ\text{C}$ to $30\text{ }^\circ\text{C}$ [95], and a pH of 7 to 9 [103], which is neutral to slightly alkaline, and promotes efficient carbon fixation via the CCM [104]. Key nutrients in the cultivation medium, such as nitrogen, phosphorus, potassium, and trace elements, are essential for optimizing photosynthesis and carbon storage in algae, ensuring a high biomass yield and effective CO_2 conversion.

4.2. Advancements in Algae-Based Carbon Capture

Optimizing photosynthetic carbon fixation efficiency is a primary objective in algae-based carbon capture. Recent research has predominantly focused on enhancing algal

traits, advancing algal cultivation techniques, optimizing cultivation systems and their parameters, and refining carbon sources to achieve this goal.

4.2.1. Enhancement of Algal Traits

Algae exhibit distinct traits that significantly influence their potential for carbon capture and biomass production. Advances in biotechnological methods have enabled the modification of these traits to improve algae's efficiency in CO₂ fixation and biomass accumulation [105,106]. Traditional approaches like selective breeding and strain development [107,108] and environmental adaptation [109,110] have long been used to improve algal traits. However, both methods typically require extended periods of time to achieve significant improvements.

Recent breakthroughs in genetic engineering, metabolic engineering, and synthetic biology offer more rapid and precise methods to enhance algal traits [106,111–113]. Yang et al. utilized genetic engineering techniques to enhance traits in the green microalga *Chlorella vulgaris*. By introducing cyanobacterial fructose-1,6-bisphosphate aldolase, guided by a plastid transit peptide, they significantly improved the algal photosynthetic capacity and cell growth. Their findings indicate that the overexpression of aldolase promotes the regeneration of ribulose-1,5-bisphosphate in the Calvin cycle and enhances energy transfer in photosystems, thereby increasing photosynthetic efficiency for carbon capture [114]. Hlavova et al. provided a comprehensive review of advancements in genetic engineering techniques for modifying microalgal traits [106]. Modern genetic engineering methods can optimize specific traits by introducing exogenous DNA into algae or through external induction, resulting in genetically modified mutants with enhanced characteristics such as increased cell density, improved CO₂ capture efficiency, and accelerated biomass production [115,116]. Brar et al. elaborated on metabolic engineering techniques, including adjusting the expression levels or activities of key enzymes in metabolic pathways or altering enzyme structures to enhance their stability and activity under specific conditions, thereby improving the efficiency of CO₂ conversion to target biomass in algae [117].

Moreover, synthetic biology, which integrates artificial genetic and metabolic engineering, offers opportunities for developing desired traits in algae [118]. CRISPR-Cas systems, one of the most promising technologies in synthetic biology, have been improved for engineering algae, particularly for enhancing their carbon capture capabilities in biofuel production by optimizing their metabolic pathways that convert CO₂ and sunlight. Algal strains such as *Chlamydomonas reinhardtii* and *Synechocystis* sp. have been modified to improve resistance to environmental stressors and boost growth rates by manipulating pathways involved in light harvesting [119–123]. Furthermore, advancements in CRISPR-based tools, combined with high-throughput screening techniques, hold great promise for enhancing the scalability of algae as sustainable biofuel producers [118].

While these technological advancements have demonstrated success in laboratory settings, further research and outdoor trials are needed to fully assess their feasibility for large-scale applications in CCUS. The continuous refinement of these methods presents a promising pathway for accelerating the integration of algal biotechnology into sustainable energy and environmental solutions.

4.2.2. Advances in Algal Cultivation Techniques

Monoculture, the practice of growing a single species of microalgae, remains the traditional and primary method of cultivation [124]. It allows for precise control and optimization of the cultivation process based on the specific characteristics of the species. While this method ensures efficient growth under optimal conditions and facilitates standardized management and operation, it is highly sensitive to environmental fluctuations and exhibits poor resistance to stress. As a result, it requires costly maintenance and is vulnerable to contamination, which can severely impact algal growth and productivity.

To address these issues, researchers are exploring mixed-culture systems, where multiple species are cultivated together [125,126]. Mixed cultures leverage species diversity [127],

enabling adaptation to varying light, temperature, and pH conditions [128]. Moreover, the coexistence of different algal species can help suppress the growth of certain pathogens or contaminants, reducing the risk of contamination. This approach not only lowers the operational and maintenance costs associated with stringent environmental conditions, but also enhances biomass production by allowing different species to complementarily utilize the nutrients in the cultivation medium, leading to improved economic benefits [129].

To fully leverage the advantages of mixed-culture systems in large-scale production, Hassanpour et al. utilized such systems to simultaneously capture CO₂ and produce various types of biomass, enhancing the economic efficiency and productivity of biomass production [126]. Specifically, they employed a gravity enrichment method to cultivate lipid-rich and carbohydrate-rich algae based on their cell density differences. The less dense lipid-rich algae float on the surface of the culture medium, while the denser starch-rich algae settle at the bottom. This simple, flexible, and cost-effective method enhances large-scale production of algae with different storage compounds, thus addressing the challenges of industrial-scale compound production.

In addition, several studies have explored the integration of algal cultivation with wastewater [130,131] and CO₂-rich waste gases [132,133] to enhance algal biomass accumulation and improve carbon capture efficiency while mitigating environmental impacts. Nutrient-rich wastewater from sources such as dairy and sugar processing contains essential nutrients like nitrogen, phosphorus, and potassium, which, if untreated, can disrupt normal water uses, including drinking, irrigation, and recreational activities [134–137]. Using this wastewater as a nutrient source for algal cultures promotes growth, increases biomass yield, and addresses the environmental challenges posed by wastewater. Hena et al. conducted a study at a dairy farm in Perlis, Malaysia, where they used four microalgae strains—*Chlorella saccharophila* UTEX 2911, *Chlamydomonas pseudococcum* UTEX 214, *Scenedesmus* sp. UTEX 1589, and *Neochloris oleoabundans* UTEX 1185. They found that the microalgae thrived in dairy wastewater, effectively removing nutrients, and leading to increased biomass production [134].

Furthermore, transporting CO₂-rich waste gases for injection into algal cultivation systems, or situating these systems near CO₂ emission sources, represents an effective strategy for optimizing carbon capture by algae and enhancing the overall efficiency of algal cultivation. Suriya Narayanan et al. conducted experiments by establishing a cultivation system for the thermo-tolerant microalga *Coelastrella* sp. F169 near an LPG-burner and cooler setup [138]. They utilized LPG-burned exhaust gas streams as the carbon source for the microalgae cultivation system. The results demonstrated that this method achieved a promisingly high net energy ratio and underscored the significance of using this strain for effective capture and utilization of CO₂-rich gas streams. Acedo et al. established open raceway ponds near the Tucson Electric Power plant in Tucson, Arizona, to cultivate the green algal species *Chlorella sorokiniana*, utilizing the flue gas from the power plant as the carbon source [139]. The amount of CO₂ introduced into the system was closely monitored over time. The results demonstrated that leveraging synergies between CO₂ emission sources and algal cultivation can not only enhance carbon capture efficiency but also support the sustainable production of algal biofuels and bioproducts.

4.2.3. Improvement in Algal Cultivation Systems

To enhance carbon capture and biomass production, improving algal cultivation systems is essential. Paul et al. reviewed recent advancements in using microalgae for carbon capture in open and closed systems [47]. For open systems, key innovations include optimizing mixing processes via computational fluid dynamics (CFD) simulations to enhance CO₂ capture by improving mixing and contact time [47]. These optimizations also facilitate uniform nutrient distribution, preventing sedimentation and cell aggregation to improve the overall efficiency and stability of microalgal cultivation. Furthermore, in open raceway ponds (ORPs), increasing their depths while integrating transparent light scattering columns (LSCs) ensures sufficient light for deeper layers, significantly enhancing

CO₂ capture efficiency and biomass production [47,140]. For closed systems, closed raceway ponds (CRWPs) with intermittent CO₂ supply enhances CO₂ capture efficiency compared to continuous supply [141]. Additionally, extending the length of the raceway pond increases microalgae–CO₂ contact time, further improving CO₂ capture efficiency [47].

Recent advancements in closed photobioreactor (PBR) systems focus on improving carbon fixation efficiency and biomass productivity. Microbubble-assisted PBRs, integrating Venturi microbubble generators and bubble columns, significantly increase the gas–liquid contact area to promote the mass transfer efficiency of CO₂, thereby boosting the overall CO₂ capture efficiency [142]. Fluid dynamics optimizations within PBRs, such as incorporating aerodynamic airfoil-shaped deflectors and three-dimensional tangential swirl plates, improve fluid flow within the reactor [143,144]. Such enhancements extend the suspension time of microalgae and increase their contact time with CO₂, ensuring sufficient light exposure, which in turn improves CO₂ capture efficiency. Additionally, CFD simulations have been employed to optimize the hydrodynamic characteristics and aeration rates within PBR systems, enhancing CO₂ capture efficiency [145,146]. Studies also focus on improving energy efficiency by utilizing renewable energy sources, such as solar collectors, to power PBR lighting systems at night [147]. This cost-effective solution meets the light cycle requirements of algal growth, thereby promoting biomass accumulation.

Additionally, advanced Internet of Things (IoT) technologies have been applied to both open and closed systems [148,149], enabling the cultivation systems to operate in a more automated and data-driven manner. By integrating sensors, these systems can continuously monitor key environmental parameters such as light intensity, temperature, pH, and dissolved oxygen concentration. The data collected is analyzed in real-time, facilitating the identification of optimal parameters. Furthermore, the connectivity features of IoT allow for remote management and operation, enabling the automatic adjustment of lighting, temperature, and nutrient supply. This remote control and automated regulation enhance operational flexibility and efficiency. In essence, this makes microalgal cultivation systems intelligent, as it can autonomously manage and improve their operational performance, greatly enhancing cultivation efficiency and leading to more efficient carbon capture.

4.2.4. Refinement of Carbon Sources

Optimizing the carbon source by refining the supply method and form of CO₂ is crucial for enhancing algal carbon capture efficiency. Gaseous CO₂, the most commonly used carbon source for algae cultivation, transfers into the cultivation medium through diffusion across the gas–liquid interface, which is influenced by mass transfer limitations. Under specific pressure and temperature conditions, Henry's Law governs the solubility of CO₂ in liquid, limiting the CO₂ available to algae [150].

To address this issue, some studies suggest adding CO₂ absorbents to the cultivation medium to enhance mass transfer for enhanced CO₂ capture efficiency [151]. While effective, this method comes with drawbacks, such as increased costs and potential contamination of the cultivation environment.

Studies have demonstrated that converting CO₂ into a bicarbonate solution, used as a carbon source in algal cultivation systems, overcomes the limitations of gas-phase CO₂ dissolution [150,152–154]. This method leverages algae's carbon concentration mechanism, utilizing the enzyme carbonic anhydrase, which converts bicarbonate (HCO₃[−]) into CO₂. In cultivation systems, higher concentrations of bicarbonate result in more CO₂ being available for photosynthesis. Yang et al. conducted a comprehensive review on the use of bicarbonate solution as a carbon source in algae cultivation systems [150]. Their study demonstrated that, in addition to significantly improving carbon fixation efficiency in algae, bicarbonate application can enhance carbon utilization by increasing the production of intracellular total sugars, exopolysaccharides, chlorophyll, and phycocyanin. This approach is particularly well-suited for utilizing non-commercial high concentrations of CO₂ from industrial processes, such as flue gas generated by power plants, steel mills, or cement factories. By dissolving CO₂ to create bicarbonate solutions for algal cultivation,

algae can capture more non-commercial CO₂, while also avoiding the safety and cost challenges associated with transporting and storing gaseous CO₂, thereby contributing more significantly to CCUS efforts.

5. Algae-Based Carbon Utilization

The utilization of carbon by algae is primarily achieved through the circular bioeconomy within the framework of CCUS. Specifically, algae capture CO₂ from the atmosphere and efficiently convert it into organic carbon compounds in algal biomass, which serves as an intermediary product. This biomass can then be further processed into biofuels and other valuable products, thereby utilizing carbon in a form that creates economic value and contributes to the CCUS process.

Initially, algae, especially macroalgae, were primarily used in food and feed due to their high protein content [37]. Over time, their potential in producing bioactive compounds like proteins, fatty acids, pigments, vitamins, and carbohydrates has been well established. These components offer exceptional antioxidant, anti-inflammatory, and free radical scavenging properties, leading to their widespread applications in food, feed, supplements, pharmaceuticals, and cosmetics [37,84,155,156].

Beyond these applications, the potential of algae in the biofuel sector garners increasing attention. Biofuels are renewable fuels produced through biological processes that convert carbon into energy. During plant growth, atmospheric CO₂ is absorbed, and when biofuels are combusted, the CO₂ released is roughly equivalent to the amount absorbed, making biofuels a relatively clean energy source [53]. Compared to traditional fossil fuels, biofuels offer key advantages, including renewability through continuous biological replenishment, a reduced environmental impact, and a wide range of energy sources, making them a more sustainable and eco-friendlier alternative. Biofuels have evolved through three generations, based on the source of feedstock [157–159]. First-generation biofuels, derived from edible plants, have been deemed unsuitable for long-term fuel production due to competition with food resources. Second-generation biofuels use non-edible plant materials or agricultural waste, which mitigates the food competition issue, but the fuel quality is generally lower. Third-generation biofuels from algae have garnered attention due to algae's rapid growth, high yield, low land use, and efficient CO₂ utilization, making them a more sustainable option. Algae efficiently absorb CO₂ through photosynthesis, converting it into energy-rich biomass for biofuel production via an organic pathway. The CO₂ released during biofuel production and combustion can be absorbed by algae again for cultivation, creating a closed-loop system that aligns with the BECCS framework and offers significant potential for net-negative carbon emissions. The basic process of algae-based carbon utilization for biofuel production under BECCS is shown in Figure 3, with more details on specific technologies provided in subsequent subsections.

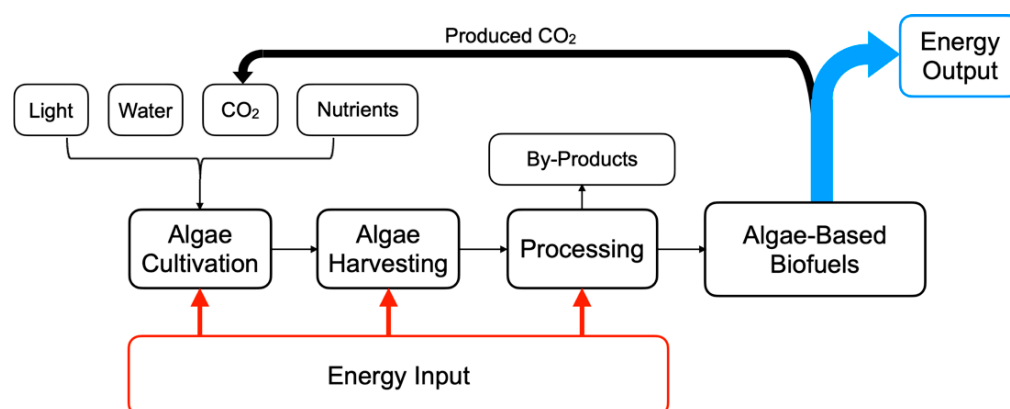


Figure 3. Schematic of biofuel production from algae.

5.1. Solid Algae-Based Biofuels

Solid algae-based biofuels are produced by processing algae into a solid form. Various components of algae, including lipids, lipid derivatives, carbohydrates, and proteins, can be effectively utilized to generate energy, thereby maximizing the energy potential of algae [48]. These biofuels are typically used through direct combustion for heating, electricity generation, and energy supply in industrial processes. The production process for solid algae-based biofuels is relatively straightforward, making it a viable option for large-scale algae utilization. It also holds potential as an alternative to wood and charcoal as fuel sources [160–162]. Currently, the most common forms of solid algae-based biofuels are solid biomass pellets (also known as bio-pellets or bio-pellet fuel) and biocoal.

5.1.1. Algae-Based Solid Biomass Pellets

Algae-based solid biomass pellets are primarily produced from microalgae through densification techniques that enhance their energy density while making their storage and transportation more efficient. These pellets offer significant advantages as a fuel, particularly due to their high efficiency during direct combustion and low carbon emissions.

The production of algae-based solid biomass pellets involves drying, crushing and screening, and pelletization. The algal biomass is typically dried using solar dryers or other drying equipment to remove moisture, improving combustion efficiency and energy output for subsequent processes [149]. The dried biomass is then crushed and screened for size consistency before undergoing pelletization, which compacts it into dense, uniform pellets, making storage and transport more efficient.

These pellets feature a small volume, high energy density, and compatibility with existing coal-fired boilers, allowing for direct combustion to maximize energy generation from algae [48,163]. Additionally, the main products of solid pellet combustion are typically CO₂ and ash, both of which can be fully recycled into algae cultivation [48]. The CO₂ serves as a carbon source for photosynthesis, while the ash, containing minerals and nutrients, can supplement the algal growth medium. While the commercialization of algae-based pellets is limited, studies have demonstrated their potential as effective biofuels. Kosowska-Golachowska et al. tested *Oscillatoria* sp. pellets in circulating fluidized bed (CFB) boilers, achieving a high heating value of 15.86 MJ/kg, fast ignition at 308 °C, and efficient combustion times [164]. Miranda et al. produced *Scenedesmus* pellets with a density of 788 kg/m³ and an energy density of 14,165 MJ/m³, which is higher than that of industrial wood-based fuels (around 11,000 MJ/m³), highlighting their significant biofuel potential [161].

Energy density and durability are critical parameters for pellet fuels, impacting combustion efficiency and costs. Optimizing pelletization through higher compression pressure [160], suitable additives [165], and elevated temperature [160] enhances the overall pellet quality. Torrefaction pelletization, which combines torrefaction with pelletization, improves biomass properties [166]. Torrefaction is a thermal-chemical treatment where biomass is heated to temperature typically between 200 °C and 300 °C in the absence of oxygen. This method benefits from torrefaction's ability to remove moisture and volatile organic compounds, making biomass with enhanced physical and chemical characteristics. This results in solid biomass pellets with higher energy density, improved durability, and better structural integrity.

Additionally, research on algae-based solid pellet biofuels focuses on reducing production costs by the co-pelletization of algae with cheaper biomass materials, such as softwood, hardwood, and rice husks [48]. For these algae-based co-pellet biofuels, the addition of small amounts of algae does not significantly increase costs but greatly improves combustion performance and durability while reducing energy consumption during pelletization.

These enhanced fuel properties are primarily attributed to the addition of algae. Their smaller particle size fills gaps between biomass particles, improving co-pellet density and durability through enhanced intermolecular forces [167]. Additionally, algae increase the biomass per unit volume, raising the energy density of the fuel and enhancing combustion

performance [167]. Furthermore, the high lipid content of algae not only increases pellet density [48] but also improves solid bridging and mechanical interlocking between particles [162], resulting in a stronger pellet structure. This structural enhancement reduces fragmentation and disintegration during combustion, further improving the durability of the co-pellets. Algae also help to lower energy consumption during pelletization, primarily due to the lubricating effects of fats and oils [165,168]. Moreover, certain components in algae, such as cellulose and hemicellulose, enhance the plasticity of the biomass mixture, facilitating pellet bonding during compression with reduced energy demand [48].

Recent studies suggest combining algae-based solid pellets or co-pellets with other fuels like coal [169], sludge [170,171], and oil shale [172], to form dual-fuel mixtures. This approach has shown several advantages. Compared to using coal alone, the co-combustion of algae-based pellets or co-pellets with coal can lower ignition and burnout temperatures, increase the combustion index, and reduce ignition delay times [48,169], thus significantly enhancing the overall combustion performance. In dual-fuel combustion with sludge, the carbon and chemical components in algae catalyze sludge residue degradation, reducing sludge melting points and accelerating combustion [170,171]. This synergistic effect not only increases combustion efficiency but also promotes complete sludge combustion. When algae-based pellets or co-pellets are co-combusted with oil shale, in addition to improving combustion efficiency, certain minerals in the oil shale react with NO_x produced during algae-based fuel combustion to form more stable compounds, thereby reducing NO_x emissions [172].

5.1.2. Algae-Based Biocoal

Algae-based biocoal is produced through the pyrolysis and carbonization of algal biomass at specific high temperatures under anaerobic conditions [173,174]. This process removes volatile organic compounds and cellulose and reduces the oxygen content. The primary components retained are fixed carbon and a small amount of ash. With its high carbon content, energy density, and stability, this solid biofuel shares characteristics similar to fossil coal, making it increasingly attractive to carbon-intensive industries [175].

Pyrolysis is a well-established technology for biocoal production from algae and other biomass [176], typically at temperatures between 300 °C and 700 °C under anaerobic or low-oxygen conditions. Drying algae through low-temperature methods or other dewatering techniques is crucial to minimize moisture content, ensuring efficient heat use and preventing disruptions in temperature distribution. However, algae with a high moisture content, such as macroalgae, require extra energy for water evaporation during the pyrolysis process, including the drying stage. This leads to reduced production efficiency and increased production costs.

To address this challenge, hydrothermal carbonization (HTC) has emerged as a promising method [173], operating under high pressure and low temperatures (typically 180 °C to 250 °C). In this process, water (referred to as “supercritical water” or “subcritical water”) acts as a reaction medium with strong dissolving capabilities. This enables organic materials in the biomass, such as cellulose, hemicellulose, and lignin, to undergo a series of chemical reactions, including hydrolysis, condensation, and rearrangement, ultimately producing biocoal as solid products along with liquid and gas by-products. In other words, HTC utilizes the moisture present in the biomass rather than expending extra energy to remove it. This characteristic is a significant advantage of the technology, as it reduces energy consumption for drying algae, improves production efficiency, and lowers production costs. The biocoal produced through HTC has properties similar to those of low-rank coal. Studies on various macroalgae using HTC have evaluated the combustion properties of the resulting biocoal, demonstrating the effectiveness of this technology in biocoal production [173]. Although HTC technology has not yet been widely adopted, its low energy consumption presents significant potential for biocoal production.

5.2. Liquid Algae-Based Biofuels

Liquid algae-based biofuels, produced through more complex chemical and biological processes than solid ones, offer higher energy densities due to their high concentrations of high-energy biomass components, particularly lipids or carbohydrates. This higher energy density makes them suitable alternatives to traditional liquid fuels, particularly in transportation and other sectors. Biodiesel and bioethanol are representative liquid algae-based biofuels. Biodiesel is produced from algal lipids, while bioethanol is derived from algal carbohydrates.

5.2.1. Algae-Based Biodiesel

Biodiesel is a fuel composed of monoalkyl esters of long-chain fatty acids derived from renewable resources [53]. Algae-based biodiesel primarily consists of Fatty Acid Methyl Esters (FAMES), produced from the lipids in algae, particularly Triacylglycerols (TAGs). While lipid content varies by algae species and cultivation conditions, it typically ranges from 20% to 50% of the dry weight [53,177].

The production process of algae-based biodiesel involves three key steps: TAGs extraction, transesterification, and separation. Extraction methods such as mechanical extraction, solvent-assisted extraction, supercritical fluid extraction, ultrasonic extraction, and biological extraction are used to disrupt the algae cells and release intracellular TAGs. In transesterification, the TAGs react with methanol in the presence of a catalyst to produce FAMES, with glycerol as a by-product, as illustrated in Figure 4. Finally, FAMES are separated from the reaction mixture. In addition to biodiesel as the primary product, glycerol, a valuable by-product, has extensive applications in cosmetics, pharmaceuticals, food additives, and chemical industries.

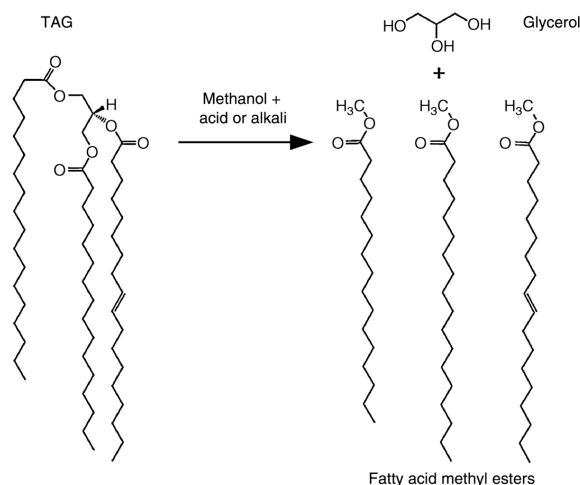


Figure 4. Production of biodiesel via transesterification of triacylglycerols (TAGs) extracted from algae. Reprinted with permission from [128]. Copyright 2010, Elsevier Ltd.

In recent years, advances in algae-based biodiesel production have focused on increasing TAG content in individual algae cells and improving transesterification catalysts. Biodiesel yield depends on both the overall biomass concentration and intracellular oil content. In addition to optimizing the cultivation environment, enhancing the intracellular TAGs content is crucial. Nitrogen limitation has been shown to effectively increase TAGs accumulation by redirecting captured CO₂ converted into storage molecules, such as TAGs or starch, rather than proteins [178]. A two-stage process has been proposed, starting with cultivation in nutrient-rich conditions for rapid biomass accumulation, followed by nitrogen limitation to induce stress and enhance intracellular TAG accumulation [179,180]. Additionally, altering the carbon-to-nitrogen ratio by adding organic carbon induces stress, boosting lipid accumulation [181]. Blocking starch synthesis under nitrogen-limited con-

ditions further redirects carbon toward TAGs, increasing lipid content and production efficiency [182].

In transesterification, catalysts are essential for enhancing efficiency and product quality [53]. Acidic catalysts offer high yields but have slower reaction rates and recovery issues [183], while basic catalysts exhibit high activity but may lead to saponification, reduced yield, and increased viscosity [53]. Dual-function acidic-basic catalysts combine the benefits of both types but may not always perform as expected [53]. Enzymatic catalysts, with their unique acyl migration mechanism, offer significant benefits in improving the transesterification reaction completeness, reducing unreacted intermediates and by-products, and enhancing biodiesel purity and yield [184].

Recent research has focused on enhancing enzymatic catalysts through advanced molecular techniques [185,186] and protein engineering [187]. These modifications include alterations to the enzyme's structural features and active sites to enhance reaction activity and substrate selectivity [188]. Additionally, enzyme immobilization on carrier materials optimizes enzyme positioning, maintains activity, and aids recovery [189]. Nanotechnology integration enhances transesterification efficiency by using nanostructured supports, such as nanofibers, nanoparticles, nanotubes, and nanopolymers, for enzyme immobilization. These supports, with high surface area and porosity, increase enzyme loading capacity, reduce mass transfer resistance, enhance mechanical strength, and improve stability and activity, thereby significantly boosting biodiesel production [190–192].

5.2.2. Algae-Based Bioethanol

Bioethanol, as a renewable energy source, is considered one of the most promising alternatives to gasoline for transportation fuel. Efficient bioethanol production requires selecting algae with high carbohydrate content [57]. Microalgae species such as *Chlorella*, *Dunaliella*, *Chlamydomonas*, and *Scenedesmus*, which contain over 50% carbohydrates, are regarded as ideal raw materials for bioethanol production [57].

Algae-based bioethanol production involves four main steps: pretreatment to break down cell walls, hydrolysis (typically enzymatic hydrolysis) to convert complex carbohydrates into simple sugars, fermentation by microorganisms to produce bioethanol, and final distillation/purification to meet fuel specifications.

Pretreatment is essential for algae-based biofuels like bioethanol, biohydrogen, and biogas [193–196]. Advancements in pretreatment technologies focus on reducing chemical use, minimizing energy consumption, and preserving sugar content. Table 2 presents commonly used pretreatment methods and their characteristics. Acid pretreatment, effective for both microalgae and macroalgae, dissolves hemicellulose and exposes cellulose fibers, thereby enhancing cellulase hydrolysis. It also degrades polysaccharides, lowers energy consumption, and increases sugar yields, making it a cost-efficient option compared to other pretreatment technologies [197].

Table 2. Algal biomass pretreatment methods and their characteristics.

Pretreatment Methods	Technical Details	Applicable Conditions	Characteristics	References	
Physical Pretreatment	Mechanical comminution	Grind, slice, and pulverize algal biomass to reduce particle sizes and cellulose crystallinity.	Effective for macroalgae with rigid cell walls and high cellulose crystallinity	Increasing the surface area-to-volume ratio of algal particles improves processing efficiency; Controlling the particle size within 3–6 mm optimizes energy consumption.	[193,198]
	Irradiation	Gamma rays disrupt algal cell walls, increase surface area, and reduce cellulose crystallinity. Microwave irradiation generates internal heat, creating an explosion effect to break down resistant cell walls.	Gamma irradiation is suitable for macroalgae with recalcitrant cellulose structures; Microwave irradiation is suitable for microalgae	Combining this method with other pretreatments significantly accelerates saccharification; High energy requirement	[199,200]
	Ultrasonication	Ultrasound generates shear forces to disrupt cell walls and reduce particle size.	Suitable for microalgae with tough cell walls and larger biomass particles	Processing efficiency is influenced by ultrasound power, temperature, and pH, requiring adjustment of treatment conditions; While it consumes high energy, its efficiency can shorten pretreatment time, ultimately lowering overall energy demand.	[201,202]
Physicochemical Pretreatment	Hydrothermal pretreatment	Hot water under high pressure hydrates cellulose and removes hemicellulose.	The most effective method for macroalgae with high cellulose and hemicellulose content	Water as the sole solvent; Short reaction time; Reduce sugar degradation loss; Minimize fermentation inhibitors	[203,204]
	Steam explosion	High-pressure steam rapidly heats and depressurizes size-reduced biomass, causing explosive expansion that ruptures algal cell walls and increases surface area.	Suitable for microalgae with high cellulose content	High energy consumption; Higher cost; The addition of chemicals can improve the conversion rate of polysaccharides to monosaccharides.	[205,206]
	Supercritical carbon dioxide	By permeating biomass with supercritical carbon dioxide, and sudden depressurization causes the rupture of cell walls and other structures.	Suitable for both of macroalgae and microalgae with high cellulose content.	Cost-effective; Easy to extract and recover; Environmentally friendly	[193]

Table 2. Cont.

Pretreatment Methods	Technical Details	Applicable Conditions	Characteristics	References	
Chemical Pretreatment	Alkaline pretreatment	Saponification is utilized by alkaline substances to break the ester bonds between hemicellulose and other components.	Suitable for both macroalgae and microalgae with high hemicellulose content	Reduce fermentation inhibition; Lower production costs.	[207,208]
	Acid pretreatment	Acidic solutions effectively disrupt algal cell walls.	Widely applicable to both microalgae and macroalgae, and the most effective method for microalgae	High sugar yield; Simple operation; By-products like furfural and hydroxymethylfurfural (HMF) may impact fermentation and require additional treatment.	[197,209]
	Sodium chlorite treatment	Sodium chlorate generates chlorine dioxide (ClO ₂) in an acid, converting lignin into soluble compounds.	Suitable for lignin-containing macroalgae	Remove lignin from biomass while maximizing the retention of carbohydrates; Significantly enhance the efficiency of sugar extraction.	[210,211]
Biological Pretreatment	Microorganisms are used to partially decompose biomass for saccharification	Different microorganisms are effective under specific conditions for macroalgae and microalgae saccharification	Low energy consumption; Safe and environmentally friendly	[193,212]	
Enzymatic Pretreatment	Enzymes are employed to target and degrade specific biomass compounds	Commonly applied to microalgae	Specific compound breakdown; Mild reaction conditions; Long processing time; High enzyme costs	[213]	

The hydrolysis process is key to sugar yield and quality, directly influencing bioethanol production. Algae-based bioethanol typically uses either separate hydrolysis and fermentation (SHF) or simultaneous saccharification and fermentation (SSF) [57]. While both enzymatic and chemical hydrolysis convert algal biomass into sugars, enzymatic hydrolysis is often preferred for its higher sugar concentration [193]. It typically uses cellulases supplemented with other enzymes, such as amyloglucosidase, α -amylase, and β -glucosidase. Enzymatic hydrolysis efficiency depends on temperature, pH, and enzyme dosage [214], with the optimal values listed in Table 3. pH control involves acid-base neutralization before hydrolysis and buffer addition during the process to maintain enzyme activity. Balancing enzyme dosage avoids a low sugar yield or high costs and inhibition [215].

Table 3. Common enzymes for enzymatic hydrolysis and their optimal reaction temperatures and pH.

Enzymes	Optimal Temperature	Optimal pH	Applicable Algae	References
Cellulase	30–45 °C	4.5–5	Macroalgae	[216]
Amyloglucosidase	20–70 °C	3.50–5.50	Microalgae	[217]
α -amylase	95–115 °C	5.0–7.5	Microalgae	[218]
β -glucosidase	less than 80 °C	4.0–6.5	Macroalgae	[219]

To improve saccharification efficiency during enzymatic hydrolysis, studies have developed cellulases with enhanced tolerance and performance [220]. Genetic engineering has modified *Trichoderma reesei* to produce cellulases with higher specific activity, that are now implemented in industrial production [221]. Efficient cellulases have also been produced from bacteria [222], fungi [223,224], and thermotolerant microorganisms [225,226]. Notably, cellulases from thermophilic and psychrophilic bacteria tolerate extreme temperatures [227,228], while marine fungus cellulases withstand high salinity and alkaline conditions [229–231].

Hydrolysis of algal biomass yields pentoses and hexoses [193]. Microorganisms vary in their ability to ferment these sugars, as detailed in Table 4. Recent research focuses on genetically engineered microorganisms capable of simultaneously fermenting both sugar types [193], enhancing bioethanol production efficiency.

Table 4. Common microorganisms for fermentation of specific sugars in algae-based bioethanol production.

Sugar Types	Fermenting Microorganisms	Characteristics of Fermenting Microorganisms	References
Pentoses (Five-Carbon Sugars)	Xylose-fermenting microorganisms, including <i>Candida shehatae</i> and <i>Pichia stipitis</i>	Specific fermentation procedures and conditions are required; Repeated batch operations can enhance bioethanol production capacity.	[193]
Hexoses (Six-Carbon Sugars)	Yeasts, with <i>Saccharomyces cerevisiae</i> as a typical example	High bioethanol tolerance; Resistance to inhibitory substances; High osmotic resistance; Efficient bioethanol production	[57]
	Bacteria, with <i>Zymomonas mobilis</i> as a typical example	Anaerobic fermentation; Rapid fermentation rate; Efficient bioethanol production; Limited tolerance to phenolic compounds	[57]

After fermentation, ethanol must be purified by distillation, which evaporates and condenses ethanol to increase its concentration and purity [232,233]. Recent research explores integrating membrane technology with distillation through membrane distillation (MD) [232]. MD uses a temperature difference across the membrane to create a vapor pressure gradient, allowing ethanol to pass through hydrophobic microporous membranes due to its volatility while retaining non-volatile substances in the feed solution. This

approach leverages the robustness and low energy consumption of membrane technology, demonstrating significant potential for commercial production.

5.2.3. Integrated Production of Algae-Based Biodiesel and Bioethanol

Biodiesel and bioethanol, as liquid algae-based biofuels, are produced through different processes using distinct primary biomass; biodiesel is derived from algal lipids, while bioethanol is produced from algal carbohydrates. To optimize the utilization of algal biomass, a practical approach is to first extract lipids from the algae to produce biodiesel. The remaining algal residues, which still contain substantial carbohydrates, can then be subjected to produce bioethanol [57]. Furthermore, after bioethanol production, the residual biomass, rich in organic matter and valuable minerals, can be utilized as a biofertilizer. This integrated approach not only enhances resource efficiency but also reduces waste production and potentially increases economic benefits, making the overall process more sustainable and economically viable.

5.3. Gaseous Algae-Based Biofuels

Although gaseous algae-based biofuels generally have lower energy density compared to solid and liquid algae-based biofuels, they typically produce fewer pollutants during use. Additionally, the transportation of gaseous biofuels is more convenient through existing pipeline infrastructure. Algae can be converted into several types of gaseous biofuels using various production technologies, including biohydrogen, biogas, and syngas.

5.3.1. Algae-Based Biohydrogen

Hydrogen gas, with a high energy density and water as its only by-product, is a clean, promising fuel for the future [59]. Biohydrogen, produced through biological processes, can be derived from algae due to their photosynthetic ability, hydrogenase activity, environmental stress tolerance, and high light conversion efficiency [234–236]. Algae-based biohydrogen production can be achieved through two primary methods: biological photolysis and dark fermentation.

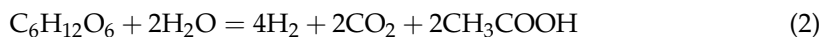
Under sufficient light conditions, algae-based biohydrogen can be produced through biological photolysis by converting light energy into chemical energy via photosynthesis [237]. This process drives the algae's photosynthetic system to split water into hydrogen gas and oxygen. In this process, Photosystem II (PSII) absorbs light energy (photons), exciting pigment molecules and splitting water into oxygen, protons (H^+), and electrons (e^-). Then, electrons are transferred through the photosynthetic electron transport chain to Photosystem I (PSI), where hydrogenases (e.g., FeFe hydrogenases) aid in reducing protons to form hydrogen gas.

Biohydrogen from algae via biological photolysis is influenced by factors, including temperature, pH, light intensity, carbon source, and nitrogen source [60]. However, the most critical factor is the sensitivity of FeFe hydrogenases to oxygen. Oxygen can irreversibly damage the active site of the hydrogenase, making oxygen tolerance a major challenge for algae-based biohydrogen production through photolysis.

To address the challenge of hydrogenase oxygen sensitivity, studies have focused on enhancing oxygen tolerance [238]. Sulfur deprivation is a key method in which algae are first cultivated with adequate sulfur, followed by sulfur removal. This reduces oxygen production in Photosystem II (PSII), creating a low-oxygen environment that supports hydrogenase activity [60]. Additionally, during electron transfer, ferredoxin (Fd) and/or nicotinamide adenine dinucleotide phosphate ($NADP^+$) may be reduced and act as an electron donor to hydrogenase or nitrogenase, facilitating the reduction of protons to molecular hydrogen (H_2) [239]. Through these mechanisms, sulfur deprivation can significantly enhance hydrogen production during biological photolysis.

Dark fermentation is a viable alternative for algae-based biohydrogen production due to its simplicity, light independence, and low energy consumption [59]. In this process, hydrogen-producing bacteria (HPB) break down algal polysaccharides, such as starch and

cellulose, under anaerobic conditions to produce hydrogen and by-products. Specifically, after pretreating algae to decompose polysaccharides into monosaccharides, HPB ferment these sugars to produce hydrogen primarily through two key metabolic pathways: the acetate pathway (Reaction (2)) and butyrate pathway (Reaction (3)) [59,240].



Several factors influence algae-based biohydrogen production during dark fermentation. Carbohydrate-rich algae are more suitable for this process than protein-rich algae, as carbohydrates provide easily fermentable sugars [59]. Excess proteins can release ammonia, inhibiting hydrogen production [241]. Sambusiti et al. summarized biohydrogen production yields from various microalgae and macroalgae, highlighting the impact of algal composition on hydrogen production [59]. Efficient pretreatment is critical to converting complex carbohydrates into simple sugars while avoiding the degradation of carbohydrates and the formation of inhibitory by-products, both of which reduce biohydrogen yield [59,242]. Additionally, optimizing HPB species, refining bacterial culture conditions, and using genetic engineering to modify metabolic pathways can enhance biohydrogen production [243].

To enhance the economic feasibility of algae-based hydrogen production, integrating dark fermentation into a biorefinery process is a promising strategy. This approach maximizes energy recovery by converting by-products into bioenergy or high-value biomolecules [59]. Common methods include: (1) integrating dark fermentation with anaerobic digestion to produce hydrogen and biogas; (2) integrating dark fermentation with photo fermentation, where purple non-sulfur bacteria convert organic acids into hydrogen under light; and (3) integrating dark fermentation with biomolecule production or algae cultivation, allowing volatile fatty acids to be converted into valuable biomolecules like polyhydroxyalkanoates (PHAs), or to support microalgae growth [244].

5.3.2. Algae-Based Biogas

Biogas is a gaseous fuel primarily composed of methane and carbon dioxide, widely used for heat generation and power production [245]. The production of biogas occurs through the anaerobic digestion of organic matter by microbial consortia, breaking down the material into combustible methane and other by-products. This process makes biogas a sustainable and renewable energy source.

In biogas production, algal biomass requires pretreatment before anaerobic digestion to release intracellular organic matter for microbial digestion. Anaerobic digestion is a complex process, and Figure 5 illustrates its four stages along with the associated chemical reactions [246,247]. In hydrolysis, hydrolytic bacteria secreting enzymes break down complex organic polymers into simpler compounds, with hydrogen being released. In acidogenesis, the hydrolysis products are fermented by acidogenic bacteria to produce alcohols, aldehydes, volatile fatty acids (VFAs), acetate, and carbon dioxide. In acetogenesis, the residuals are further converted into acetate, carbon dioxide, and hydrogen. Finally, in methanogenesis, methanogenic archaea convert acetate, carbon dioxide, and hydrogen into the target product, biogas.

The final yield of algae-based biogas depends on the synergistic activity of microorganisms in anaerobic digestion. Optimizing microorganisms' growth parameters is crucial for enhancing anaerobic digestion efficiency and biogas production. Kamusoko et al. reviewed several key operational factors [246], including the pH [248], temperature [246], carbon-to-nitrogen ratio [249], agitation [250], inoculum [251], and other factors [252–254], and suggested measures to enhance biogas yields. Additionally, the remaining digestate can be used as fertilizer, improving soil quality, and increasing the overall economic value of algae-based biogas.

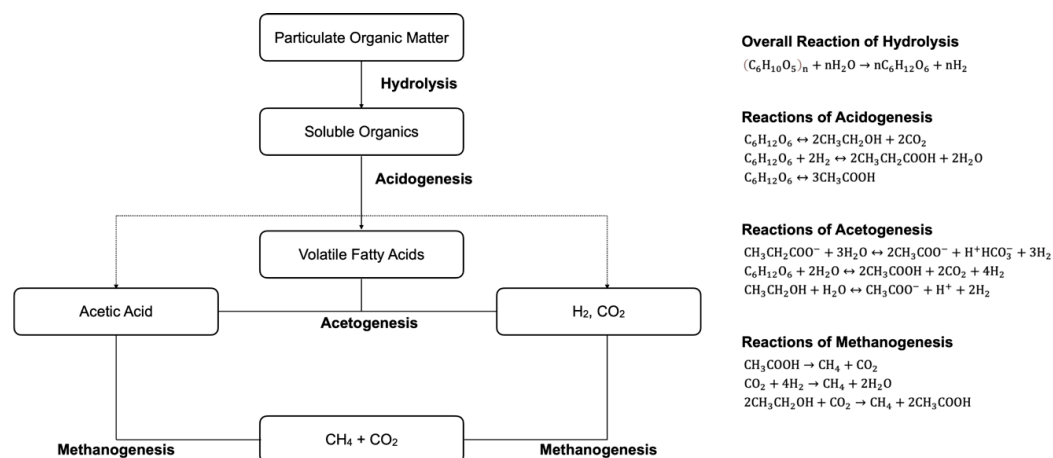


Figure 5. Schematic of algae-based biogas production through anaerobic digestion. Reproduced with permission from [246]. Copyright 2022 Elsevier Inc.

5.3.3. Algae-Based Syngas

Syngas is a gaseous mixture of hydrogen, carbon monoxide, carbon dioxide, and methane, with small amounts of steam and other by-products [255,256]. It has broad applications, including use as fuel for power generation and an intermediate for producing other fuels, like diesel, methanol, and hydrogen.

Algae-based syngas is produced through gasification, a process where biomass is converted into syngas at high temperatures (700–1000 °C) with oxidants like oxygen, air, and steam. The process involves drying, pyrolysis, combustion, and reduction [255]. Initially, the algal biomass is dried to remove moisture. In pyrolysis, biomass is decomposed at high temperatures into char, tar, and volatiles. In the subsequent combustion and reduction phases, these pyrolysis products, which are primarily carbon-based, react with oxidants (Reactions (4)–(6)) and undergo additional chemical reactions among themselves (Reactions (7)–(11)) [255], resulting in the production of syngas.



The production of algae-based syngas is influenced by the algae type and characteristics, as well as by gasification operational parameters, such as temperature, pressure, and airflow rate. Studies using modeling [257] or simulation software [255,258,259] have conducted detailed analyses of different gasification stages. Optimizing process parameters through simulation lowers production costs and improves both the efficiency and quality of syngas production in practical applications.

Research also focuses on enhancing pyrolysis technology to increase the yield, quality, and characteristics of syngas. Lee et al. reviewed traditional and advanced pyrolysis techniques, evaluating the advantages and disadvantages of each method [52]. They highlight catalytic pyrolysis [260], co-pyrolysis [261–263], hydropyrolysis [264], and microwave-assisted pyrolysis [265,266], as more efficient in converting algal biomass into syngas while

reducing waste and energy consumption. Additionally, Molino et al. provided a thorough review of technological advancements in gasification reactors for syngas production and syngas utilization for synthesizing high-energy-density biofuels, such as methanol, ethanol, dimethyl ether, hydrogen, and synthetic natural gas [256].

A summary of the typical algae-based biofuels, their primary production technologies, and their characteristics is provided in Table 5. Algae-based biofuel production technologies can be categorized into two pathways: thermochemical and biochemical. The thermochemical pathway, which includes pyrolysis, carbonization, gasification, and pelletization, involves the thermal decomposition of algal organic compounds to produce biofuels. The biochemical pathway, which includes transesterification, fermentation, biological photolysis, dark fermentation, and anaerobic digestion, primarily relies on microorganisms and enzymatic processes to convert algal biomass into biofuels. Raheem et al. have conducted a comprehensive review comparing the energy conversion efficiency and techno-economic feasibility of various algae-based biofuel production techniques [267]. While energy conversion efficiency and economic feasibility depend on multiple factors, thermochemical technologies generally demonstrate higher energy conversion efficiencies and greater economic feasibility compared to biochemical processes [267]. Although production technologies for various algae-based biofuels are continuously advancing, algae-based biodiesel is currently the one closest to commercialization. Other algae-based biofuels remain in the research or pilot stages and have not yet achieved large-scale commercial production.

Table 5. Typical algae-based biofuels with their production technologies and characteristics.

Algae-Based Biofuels	Production Technology		Characteristics
Solid Algae-Based Biofuels	Solid biomass pellet	Pelletization	Efficient utilization of algal biomass; A straightforward manufacturing process; Convenient use for direct combustion
	Biocoal	Pyrolysis and carbonization	Utilization of carbon-based components of algal biomass; Similar characteristics to coal; Applicable to carbon-intensive industries
Liquid Algae-Based Biofuels	Biodiesel	Transesterification	Utilization of algal lipid components; High energy density; Glycerol as a valuable by-product with extensive application
	Bioethanol	Fermentation	Utilization of algal carbohydrates; High energy density; Promising transportation fuel alternatives to gasoline
Gaseous Algae-Based Biofuels	Biohydrogen	Biological photolysis and dark fermentation	Utilization of algal carbohydrates, lipids, and proteins; Water as the only by-product when used as fuel; Integration with biorefinery for enhanced energy recovery and by-product utilization
	Biogas	Anaerobic digestion	Utilization of algal carbohydrates, lipids, and proteins; Digestate as fertilizer by-product
	Syngas	Gasification	Utilization of algal carbohydrates, lipids, and proteins; Direct use as fuel, or further converting into other fuels

6. Algae-Based Carbon Storage

The previous sections detail the organic pathway of algae for CO₂ capture and utilization within the BECCS framework. Algae capture atmospheric CO₂ through photosynthesis, converting it into organic biomass, which is then used to produce high-value bioenergy. This process effectively transforms atmospheric CO₂ into organic carbon within the algal cells, which is ultimately utilized as bioenergy. However, this organic pathway primarily emphasizes carbon utilization rather than carbon storage, as the captured carbon is stored only short-term before being converted into biofuels for utilization rather than being permanently stored. Recent studies have explored the potential of algae for long-term CO₂ storage.

In this section, a novel inorganic pathway is introduced, leveraging the microbial characteristics of algae to facilitate microbially induced calcium carbonate precipitation (MICP). This mechanism enables algae to store CO₂ in the long term, in the form of stable solid carbonates, distinguishing it from the previous discussed organic pathway. Through both the organic and inorganic pathways, algae offer significant advantages for further applications in CCUS.

6.1. MICP and Photosynthetic MICP for Carbon Storage

MICP is a biogeochemical process in which certain microorganisms promote the precipitation of calcium carbonate (CaCO₃) through their metabolic activities [268]. In this process, some microorganisms, such as urease-producing bacteria, catalyze the hydrolysis of urea, producing ammonia (NH₃) and carbon dioxide (CO₂). The CO₂ subsequently reacts with water to form carbonic acid (H₂CO₃), which further dissociates into hydrogen ions (H⁺) and bicarbonate ions (HCO₃[−]). These bicarbonate ions then react with calcium ions (Ca²⁺) in the environment to form calcium carbonate precipitates.

MICP was initially discovered in soil matrices and has been used as a key mechanism for various applications, particularly for soil and concrete reinforcement. By promoting the precipitation of calcium carbonates to fill soil cracks or repairing concrete, MICP enhances the strength and durability of these materials [269]. In a study by Sharma et al. on the strength enhancement and lead immobilization of sand, a consortium of *Bacillus sphaericus* and *Nostoc commune* (a blue-green algae) was employed to leverage the MICP mechanism [270]. The results demonstrated that these microorganisms catalyzed the formation of calcite (CaCO₃) crystals, which bonded sand particles together, significantly increasing the soil strength and reducing permeability. Furthermore, under optimized nutrient and chemical conditions, the biocementation process was enhanced. This study not only demonstrated that algae could improve soil strength and immobilize lead in contaminated sand via MICP, but also inspired further exploration of algae's potential for carbon capture and storage (CCS) by sequestering CO₂ in the form of stable inorganic calcium carbonate. In this inorganic pathway, MICP facilitates long-term carbon storage through the mineralization of CO₂ into a stable solid.

In algae and other photosynthetic microorganisms, particularly cyanobacteria and microalgae, these organisms can serve as biological carriers for calcium carbonate precipitation, thus facilitating the MICP mechanism driven by microbial metabolism. In algae, this mechanism can work synergistically with photosynthesis, creating a dual pathway for carbon capture and storage. During photosynthesis, algae absorb CO₂ from the environment, converting it into organic biomass. Simultaneously, the MICP process involves key microbial processes, such as ureolysis (urea hydrolysis), sulfate reduction, and denitrification, which not only elevate the pH within the microbial system, but also increase the concentration of bicarbonate ions [271]. This promotes the reaction between captured carbon and calcium ions (Ca²⁺), leading to the formation of calcium carbonate precipitates. Consequently, in algae, carbon is stored not only in the form of organic biomass through photosynthesis, but also as inorganic calcium carbonate precipitates via MICP. This creates a dual carbon storage mechanism, as illustrated in Figure 6, which involves both photosynthesis and MICP, further enhancing the potential of algae-based systems in CCUS.

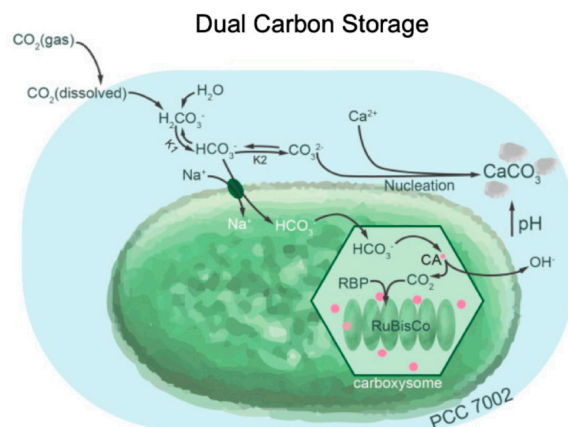


Figure 6. Schematic of CO₂ storage in Cyanobacteria PCC 7002 through biomass accumulation and MICP during photosynthesis. Reproduced with permission from [272] under the CC BY-ND 4.0 license.

The integration of photosynthesis with MICP, referred to as photosynthetic MICP or photosynthetically driven MICP, offers several advantages. Firstly, photosynthesis acts as a natural and efficient mechanism for CO₂ fixation into organic compounds, while simultaneously supplying the necessary carbon source required for the MICP process. Additionally, the energy derived from photosynthesis directly supports the metabolic reactions involved in MICP, thereby reducing reliance on external energy sources. Furthermore, compared with other industrial carbon capture and storage techniques, photosynthetic MICP is more environmentally friendly and cost-effective. Some traditional carbon capture technologies, such as amine-based CO₂ capture systems and post-combustion carbon capture, generate harmful by-products like amine degradation products or corrosive compounds, which require higher costs for additional treatment processes, chemicals, and extra energy. Some traditional carbon storage techniques, such as geological storage, can incur high costs and involve risks of leakage. In contrast, photosynthetic MICP leverages natural biological processes that do not require external energy or produce toxic by-products. This makes it a more sustainable, eco-friendly, and cost-effective choice for carbon sequestration.

Therefore, this dual carbon mechanism creates a more resilient and efficient approach for carbon sequestration, effectively addressing the need for both short-term (organic) and more importantly, long-term (inorganic) carbon storage solutions. As a result, this dual carbon storage approach has garnered significant attention in recent carbon sequestration research. Advances in this field are primarily focused on utilizing algae as core materials, combined with other advanced technologies, to achieve more efficient and long-term CO₂ storage.

6.2. Dual Carbon Storage Using 3D-Printed Structures Embedded with Living Algae

The algae-based photosynthetic MICP dual carbon storage mechanism shows significant potential for practical and widespread application due to its scalability. Research has proposed developing scalable engineered photosynthetic MICP living materials for carbon storage infrastructure, further enhancing its applicability.

Dranseike et al. proposed a method that integrates biomanufacturing techniques with living algal materials to achieve dual carbon storage through photosynthesis and MICP [272]. In this study, Pluronic F-127 (F127), a biologically inert transparent material, was selected as the matrix for algal growth. The transparency of this hydrogel ensures sufficient light penetration, supporting the normal growth and photosynthetic activity of the encapsulated algae. Additive manufacturing techniques were used to fabricate the matrix into intricate open lattice structures, branched forms, and independent columnar shapes, inspired by aquatic structures and capillary-driven flow [273–275]. Figure 7 shows 3D-printed lattice structures that facilitate light exposure and medium transport through passive capillary wetting, enabling the growth of algae via photosynthesis. The application of 3D printing technology enabled the precise fabrication of these complex struc-

tures [101,276,277], optimizing the matrix design to enhance light exposure and nutrient exchange for the algae. As a result, the study demonstrated that the algae in these optimized bioprinted structures could survive and grow normally for over a year, successfully achieving dual carbon storage through biomass production via photosynthesis and the formation of insoluble carbonates via MICP. Furthermore, the mineral phase of the insoluble carbonates produced by MICP mechanically strengthened the bioprinted structures. This research highlights the potential of combining algal characteristics with biomanufacturing techniques to harness the dual carbon storage capabilities of algae, offering a scalable solution for CO₂ storage.

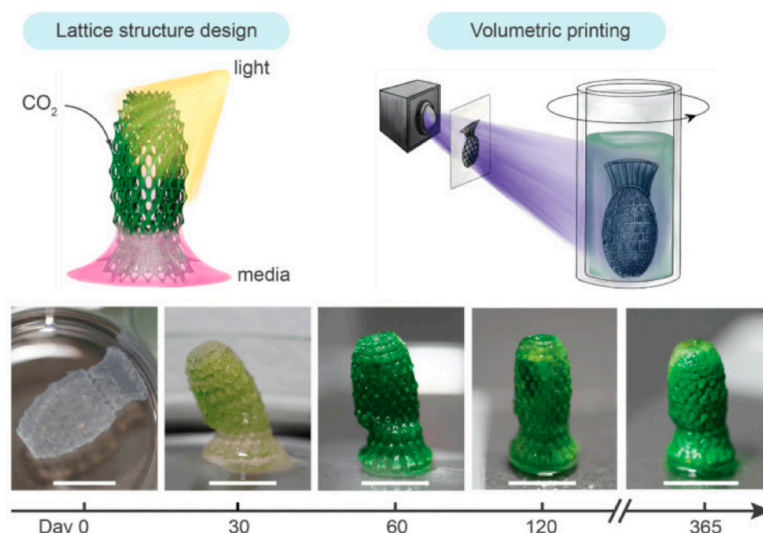


Figure 7. Digitally designed and volumetric-printed lattice structures facilitating light exposure and medium transport, enabling the growth of algae via photosynthesis. Reproduced with permission from [272] under the CC BY-ND 4.0 license.

As CCUS research increasingly shifts toward biologically driven processes due to their sustainability, eco-friendliness, and cost-effectiveness, the integration of biomanufacturing techniques with algae-based systems could guide future advancements. The scalability of such systems, particularly with the development of additive manufacturing, enables the creation of high-resolution structures optimized for specific algae cultivation conditions and adaptable to various environments. Future research could focus on selecting appropriate algae species for specific environments, such as those with low-temperature tolerance for cold regions, optimizing additive-manufactured structures to ensure algae survival and longevity, and applying these systems for long-term CO₂ sequestration in these regions.

7. Challenges and Future Perspectives

Given the urgent need for CO₂ emission reductions and the significant potential of algae-based CCUS in both carbon mitigation and bioenergy production, many companies and research institutions worldwide are actively pursuing its large-scale commercial application. However, despite recent technological advancements that have supported the growth of algae-based CCUS, economic viability remains the primary barrier to widespread adoption. To overcome this challenge, future research should focus on three key areas: improving CO₂ capture efficiency while reducing cultivation costs; enhancing biofuel production efficiency and yield with lower energy consumption; and optimizing resource utilization while increasing the value of by-products. These strategies will strengthen the economic feasibility of algae-based CCUS and accelerate its broader implementation.

Currently, algae-based carbon capture technologies have significantly improved CO₂ capture efficiency through photosynthesis by enhancing open and closed cultivation systems and optimizing relevant cultivation parameters. Future research should focus on

optimizing algal characteristics through bioengineering and biotechnology. For instance, modifying photosynthetic pigments [278], altering photosynthesis-related proteins [279], and improving the activity and stability of carbon fixation enzymes [280] can help algae maintain efficient carbon capture under diverse cultivation conditions. Additionally, enhancing algae's resistance to bacterial and other biological contamination can significantly reduce the maintenance costs of cultivation systems. These advancements will greatly increase the efficiency of algae-based carbon capture and establish a foundation for its economic viability.

In algae-based carbon utilization for biofuel production, improving the efficiency and yield of biofuel production can be achieved by selecting suitable algal strains as feedstock. It is crucial to choose strains that can be stably cultivated and efficiently produce biomass under the specific climate conditions of the target location. Additionally, genetic and metabolic engineering can be employed to optimize specific phenotypes and metabolic pathways in algae, maximizing biomass yield and significantly improving biofuel production efficiency and output. Another key future challenge lies in improving energy conversion efficiency through the algae-to-biofuel process. The energy required for cultivation, harvesting, and conversion must be carefully managed to ensure that the net energy balance remains positive, that is, the energy output from the final biofuel exceeds the energy input. This challenge is particularly critical for the thermochemical and biochemical conversion processes, where high energy demands can offset the benefits of biofuel production. To enhance the economic feasibility of algae-based biofuels, future technological advancements should focus on optimizing cultivation systems, increasing conversion efficiency, and incorporating energy-saving techniques such as waste heat recovery. Furthermore, integrating low-cost renewable energy sources, such as solar or wind power, as primary energy inputs in the production process offers a feasible approach to lowering energy costs. These measures, combined with innovations in process efficiency, will be crucial in overcoming current economic barriers and driving large-scale commercialization of algae-based biofuels within the CCUS framework.

In the process of utilizing algae for CO₂ capture and subsequent biofuel production, it is essential to make full use of nutrient-rich wastewater and CO₂-rich exhaust gases from other industries to enhance resource efficiency. Additionally, by converting by-products or residual algae from biofuel production into high-value products such as biofertilizers, animal feed, or raw materials for the chemical industry, a diversified revenue stream can be created. This approach improves overall economic viability, promoting the sustainable development of algae-based CCUS.

8. Conclusions

Algae, with its high photosynthetic efficiency, fast growth rates, and lack of competition with arable land, has emerged as a promising candidate within the BECCS framework to address the dual challenges of reducing atmospheric CO₂ concentrations and producing sustainable bioenergy. This review provides a comprehensive overview of the advancements in algae-based CCUS technologies.

Algae, through photosynthesis, can effectively capture CO₂. In recent years, advancements in enhancing algal traits, improving cultivation techniques, optimizing cultivation systems and their parameters, and refining carbon sources have significantly increased the efficiency of carbon capture by algae.

The CO₂ captured by algae is converted into algal biomass, which can be utilized for the production of various biofuels, including solid pellets and biocoal as solid biofuels, biodiesel and bioethanol as liquid biofuels, and biohydrogen, biogas, and syngas as gaseous biofuels. Recent advancements in these technologies have led to increased yields of biofuels.

Algae, in addition to storing CO₂ as organic matter through photosynthesis, can leverage their microbial properties to store CO₂ in the form of inorganic carbon precipitation through MICP, thereby exhibiting a dual carbon storage mechanism. This dual carbon storage mecha-

nism, when integrated with additive manufacturing technology, enables the production of materials for carbon storage, facilitating scalable and feasible CO₂ storage solutions.

Economic viability remains the primary barrier to the large-scale industrial adoption of algae-based CCUS. To address this challenge, future research should focus on enhancing the efficiency of CO₂ capture by algae while reducing cultivation costs, improving the production efficiency and yield of algal-based biofuels, and minimizing energy consumption during production. Additionally, optimizing resource utilization and increasing the value of by-products will be essential to strengthen the economic feasibility of algae-based CCUS, thereby facilitating its broader application.

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